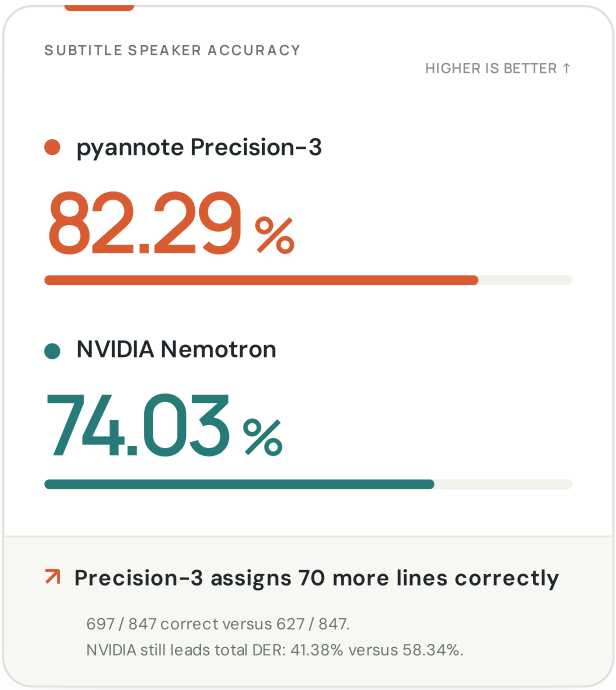
checksub.

SPEAKER DIARIZATION / UPDATED 2 NOV 2026

Speaker detection, measured.

Which model assigns a subtitle to the right person? Precision-3 and NVIDIA Nemotron, measured on the same 847 reviewed single-speaker lines across five English recordings.



THE TEST SET

Same audio, both detectors

05
Completed English recordings

40m 35s
Total audio evaluated

847
Single-speaker subtitle lines scored

4 / 5
Clips favor Precision-3 on line accuracy

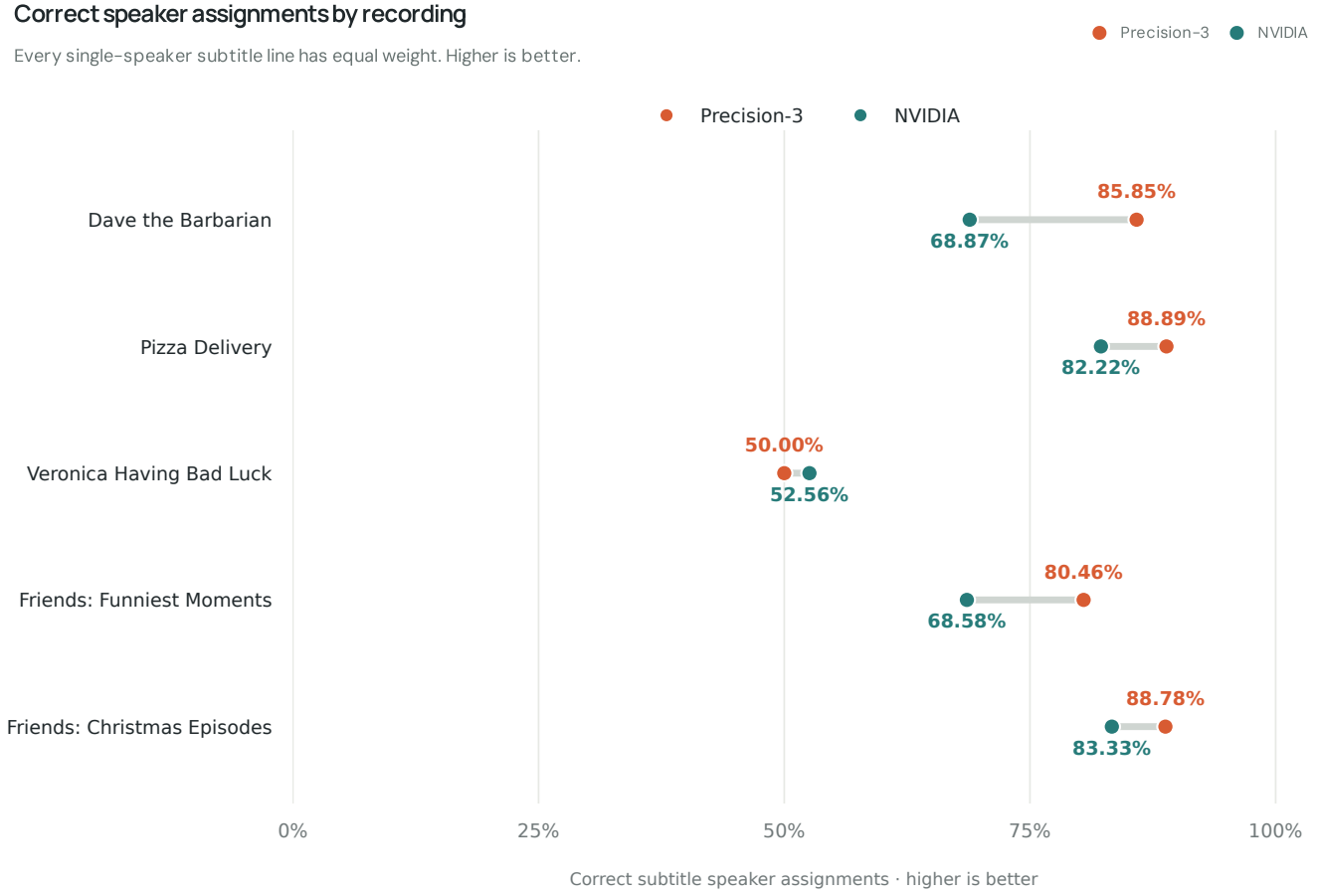
01 THE CHECKSUB ASSIGNMENT TEST

The right person. Measured per subtitle.

Keep the same subtitle timings and reviewed speaker names. Apply Checksub's current assignment rule to each model's saved output, then count correct lines. Precision-3 has 150 incorrect assignments; NVIDIA has 220.

Correct speaker assignments by recording

Every single-speaker subtitle line has equal weight. Higher is better.



847 scored lines out of 849; two five-person chorus lines assessed separately in the overlap analysis

One consistent cluster-to-name mapping per model and recording

Correct lines and incorrect assignments under Checksub's current rule

RECORDING	SCORED LINES	PRECISION-3 CORRECT	NVIDIA CORRECT	INCORRECT P3 / NVIDIA
Dave the Barbarian	106	91 · 85.85%	73 · 68.87%	15 / 33
Pizza Delivery	90	80 · 88.89%	74 · 82.22%	10 / 16
Veronica Having Bad Luck	78	39 · 50.00%	41 · 52.56%	39 / 37
Friends: Funniest Moments	261	210 · 80.46%	179 · 68.58%	51 / 82
Friends: Christmas Episodes	312	277 · 88.78%	260 · 83.33%	35 / 52
All five recordings	**847**	**697 · 82.29%**	**627 · 74.03%**	**150 / 220**

THE SAME RESULT WITHOUT CHRISTMAS

Precision-3 leads the four-clip aggregate.

420 / 535 lines correct versus 367 / 535. On the three recordings within NVIDIA's eight-speaker limit, the figures are 76.69% versus 68.53%.

Precision-3

78.50%

Excluding Friends Christmas

NVIDIA

68.60%

Excluding Friends Christmas

Assignment rule, speaker mapping and fallback counts +

The existing Checksub rule

One identity mapping per recording

Default assignments are visible

Choose the single detected segment with the largest overlap with the subtitle. The first segment wins ties. This reproduces the editor rule; overlap from separate segments of the same speaker is not summed.

All 849 intervals match the saved Checksub exports. The scorer matched the original Ruby assignment code on all 1,694 scored model/line combinations.

Each anonymous model cluster is matched to at most one reviewed person, and each person to at most one cluster. The global mapping maximizes correct line count; it is never changed per line. This measures best-case grouping agreement after consistent speaker naming.

Two Dave chorus lines have five reference speakers each. They are excluded from this single-speaker score and remain included in DER and overlap evaluation.

When no segment overlaps, Checksub defaults to speaker ID 1. Precision-3 used this fallback on 15 lines, with two correct; NVIDIA on 69 lines, with 27 correct. The main score includes this editor behavior.

Counting correct assignments only when supported by a detected segment gives **82.05% for Precision-3 and 70.84% for NVIDIA**, over the same 847 lines. Incorrect-line counts are a correction proxy, not measured editing time.

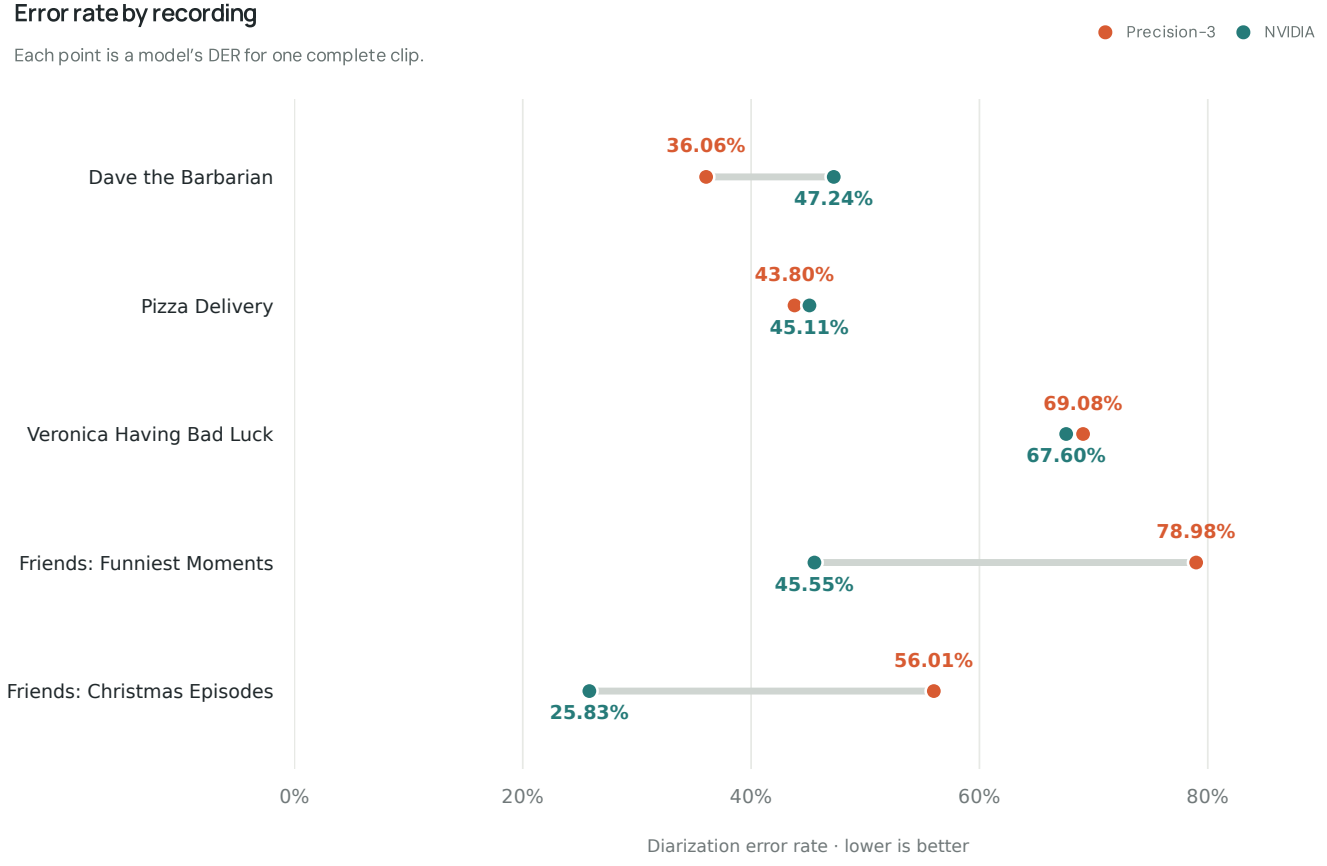
02 DIARIZATION ERROR RATE

NVIDIA leads on DER. Precision-3 on speaker confusion.

NVIDIA has lower weighted DER across the five clips. Precision-3 has less wrong-speaker time in every clip. DER scores full speech intervals; the line metric above scores the speaker assigned to each subtitle.

Error rate by recording

Each point is a model's DER for one complete clip.



0-second collar · overlapping speech included

DER is an error rate, not a percentage accuracy score.

WITHIN THE EIGHT-SPEAKER LIMIT

The three clips within the limit.

Pizza, Veronica and Friends: Funniest Moments have at most eight reference speakers. NVIDIA also leads DER on this subset, by **17.12 percentage points**. Subtitle timing limits still apply.

Precision-3

67.71%

Three recordings · weighted DER

NVIDIA

50.59%

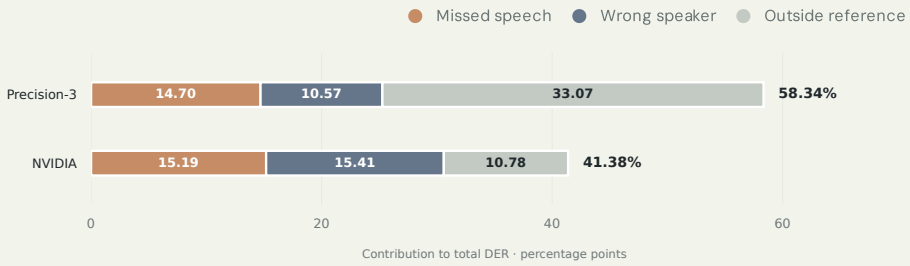
Three recordings · weighted DER

LOOKING INSIDE THE SCORE

What went wrong?

Precision-3 totals **139.68 seconds** of wrong-speaker time; NVIDIA totals **203.66 seconds**. NVIDIA's lower DER comes mainly from fewer detections outside the reference. Those intervals can contain real speech omitted from subtitles and need listening review.

How we calculate DER ↗



03 THE RECORDINGS

Every recording tells a different story.

Counts can be too low, too high or exactly right while individual assignments remain wrong. The brief ninth character in Friends Christmas contributes only 2.12 seconds of reference speech.

01 — 4:23 · 11 reference voices

Dave the Barbarian

Singing, music and a large cast

Reference caveat

Reviewed from Checksub's original subtitle export and speaker assignments. The initial speaker labels may influence human review. Cue times came from the shared Checksub transcription/alignment pipeline; those times do not inherently favor either diarizer.

Correct subtitle lines

Precision-3 85.85% NVIDIA 68.87%

Precision-3 DER	36.06%
NVIDIA DER	47.24%
Speaker labels Reference / Precision-3 / NVIDIA	11 / 9 / 5

Precision-3 lower DER

02 — 4:04 · 5 reference voices

Pizza Delivery

Rapid dialogue and background music

Reference caveat

Reviewed from Checksub's original subtitle export and speaker assignments. The initial speaker labels may influence human review. Cue times came from the shared Checksub transcription/alignment pipeline; those times do not inherently favor either diarizer.

Correct subtitle lines

Precision-3 88.89% NVIDIA 82.22%

Precision-3 DER	43.80%
NVIDIA DER	45.11%
Speaker labels Reference / Precision-3 / NVIDIA	5 / 4 / 3

Precision-3 lower DER

03 — 3:59 · 8 reference voices

Veronica Having Bad Luck

Short turns and similar character voices

Reference caveat

Reviewed from Checksub's original subtitle export and speaker assignments. The initial speaker labels may influence human review. Cue times came from the shared Checksub transcription/alignment pipeline; those times do not inherently favor either diarizer.

Correct subtitle lines

Precision-3 50.00% NVIDIA 52.56%

Precision-3 DER	69.08%
NVIDIA DER	67.60%
Speaker labels Reference / Precision-3 / NVIDIA	8 / 5 / 5

NVIDIA lower DER

04

14:01 · 8 reference voices

Friends: Funniest Moments

Live-action dialogue, laughter and long subtitle gaps

Reference caveat

Reviewed from Checksub's original subtitle export and speaker assignments. The initial speaker labels may influence human review. Cue times came from the shared Checksub transcription/alignment pipeline; those times do not inherently favor either diarizer.

Correct subtitle lines

Precision-3 80.46% NVIDIA 68.58%

Precision-3 DER 78.98%

NVIDIA DER 45.55%

Speaker labels
Reference / Precision-3 / NVIDIA 8 / 9 / 8

NVIDIA lower DER

05

14:08 · 9 reference voices

Friends: Christmas Episodes

Short interjections, laughter and a brief ninth speaker

Reference caveat

Reviewed from an SRT prepared with NVIDIA speaker labels and existing Checksub cue times. The initial labels may favor NVIDIA's measured identity agreement through review anchoring. The cue times were inherited from Checksub, shared by both models and were not independently annotated.

Correct subtitle lines

Precision-3 88.78% NVIDIA 83.33%

Precision-3 DER 56.01%

NVIDIA DER 25.83%

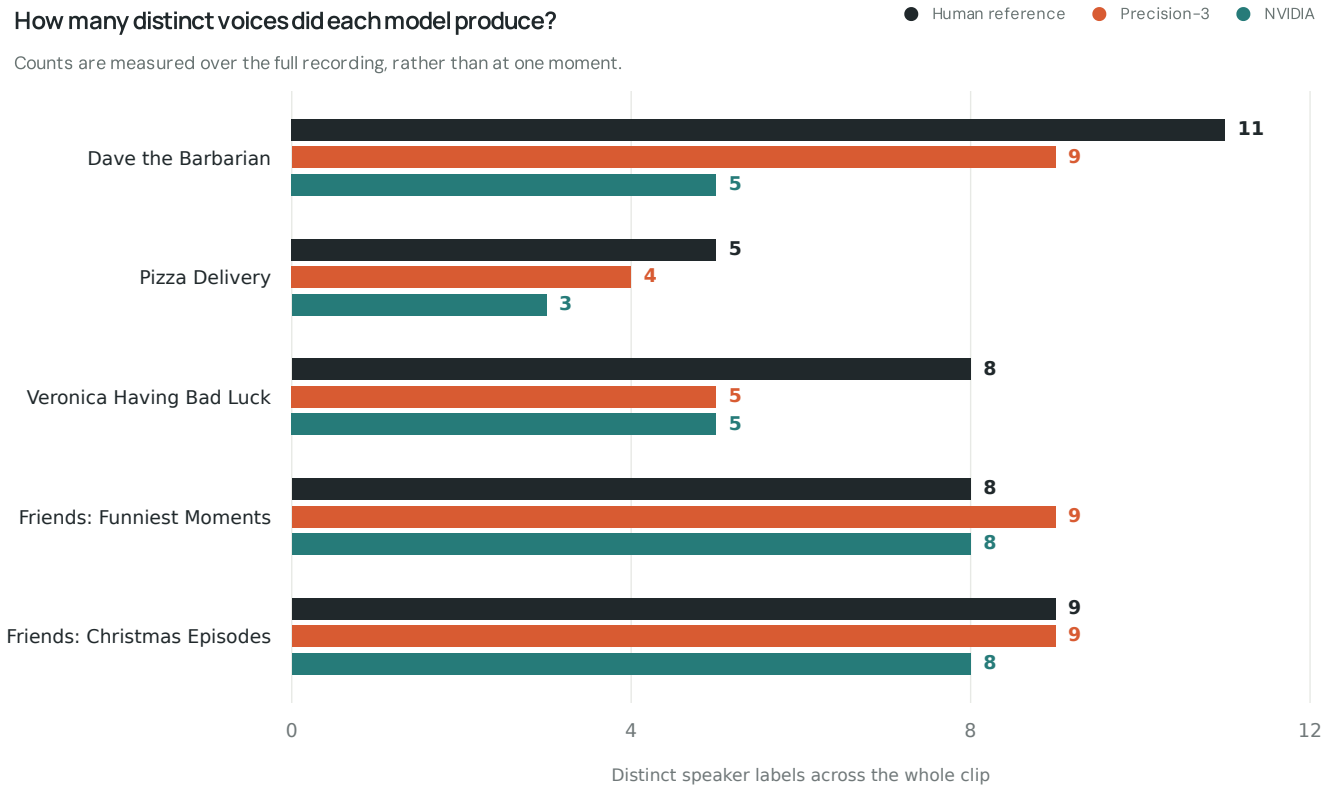
Speaker labels
Reference / Precision-3 / NVIDIA 9 / 9 / 8

NVIDIA lower DER

A material limit on this comparison: Dave has 11 reference speakers and Friends Christmas has nine. NVIDIA supports at most eight speakers per session. Both recordings exceed its capacity; the separate three-recording subset stays within the limit.

How many distinct voices did each model produce?

Counts are measured over the full recording, rather than at one moment.



04 AN INTERACTIVE CLOSER LOOK

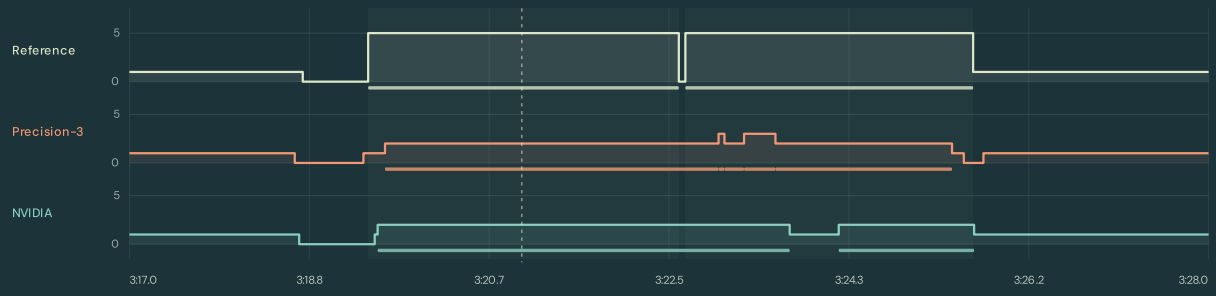
Inside the overlap.

Both models flag simultaneous voices. Explore exactly when they do, and how many speaker labels are active at the same moment.

DAVE THE BARBARIAN / FIVE-VOICE CHORUS

Five voices in the reference. Fewer in the predictions.

Human reference · Precision-3 · NVIDIA · Height = simultaneous speaker labels



3:21.00 Reference 5 Precision-3 2 NVIDIA 2

Annotated overlap

6.10s

Two chorus passages, with five reference characters singing together.

Precision-3 · within the chorus

5.71s

93.7% of annotated overlap marked with ≥2 labels. This does not verify speaker identities.

NVIDIA · within the chorus

5.50s

90.2% of annotated overlap marked with ≥2 labels. This does not verify speaker identities.

When was overlap detected?

3 intervals

TIME WINDOW	REFERENCE PEAK	PRECISION-3 PEAK	NVIDIA PEAK	REFERENCE STATUS
1:08.96–1:08.97 ↗	1	1	2	No overlap annotation
3:19.43–3:22.60 ↗	5	2	2	Annotated overlap
3:22.67–3:25.60 ↗	5	3	2	Annotated overlap

The two annotated intervals are 3:19.43–3:22.60 and 3:22.67–3:25.60. Candy, Mrs. Bogmelon, Oswidge, Faffy and Fang each have a reference row. Model-only flags outside those intervals are not independently verified. Event rows are clickable and show peak simultaneous counts inside each interval.

Detecting overlap and recovering every voice are separate tasks. In the annotated chorus, Precision-3 reaches three simultaneous labels; NVIDIA reaches two. The reference contains five. Those counts do not verify which characters were assigned correctly.

A useful pilot.
A specific set of limits.

These are our measured results, not the vendors' published benchmark scores. They describe this test set and these references.

THE METRIC

Diarization error rate

$$\frac{\text{Missed speech + extra speech + wrong-speaker time}}{\text{Reference speaker time}}$$

Multiplied by 100. **Lower is better.** DER can exceed 100%; subtracting it from 100 does not give a reliable "accuracy" score.

Scores are weighted by reference speaker time. Overlapping reference voices each contribute speaker time.

01 **Identical audio, isolated detection**

Both models received the same 16 kHz mono WAV for each clip. This compares Precision-3 with Nemotron 3 Diarization, not the full Nemotron 3.5 Transcribe pipeline.

02 **Model-assisted labels, subtitle-based times**

The first four references began with Checksub's original speaker assignments; Friends Christmas began with NVIDIA speaker labels. Those initial labels may influence human review. All five use the same Checksub transcription/alignment cue times for both models; those timings do not inherently favor either diarizer. Subtitle display boundaries can differ from audible speech. Human review does not make this a blind reference. Only Dave's chorus has explicit overlap annotation.

03 **Strict, consistent scoring**

Zero boundary collar; overlap included; full-clip evaluation. Speaker IDs are matched to names globally and one-to-one, so splitting or merging identities is penalized.

04 **A small, specialized sample**

Three animation clips and two Friends compilations, all in English, with music, laughter and short turns. Dave and Friends Christmas exceed NVIDIA's eight-speaker capacity. Customer recordings and other languages need separate evaluation.

05 **Speed and cost need a separate test**

Precision-3 used a hosted API; NVIDIA used local GPU inference. Queueing, upload and model load have different scopes. This report makes no apples-to-apples speed or cost claim.

06 CONCLUSION

Precision-3 leads on subtitle assignments.

Using Checksub's current rule and one consistent identity mapping per recording, Precision-3 assigns **697 of 847 lines correctly (82.29%)**; NVIDIA assigns **627 (74.03%)**. That is 70 fewer incorrect line assignments. Precision-3 leads this metric in four of the five recordings.

This supports keeping Precision-3 as the provisional default for the current subtitle workflow. NVIDIA wins total DER, 41.38% versus 58.34%. Both findings are valid for this test: one measures the label on each subtitle, the other measures full speech intervals. Reference review was model-assisted, and this pilot does not measure editing time or production cost. Neither system recovers all five annotated chorus voices.

● PROVISIONAL RECOMMENDATION FROM MEASURED LINE ACCURACY

BEFORE A REPLACEMENT DECISION

1 **Audit the disagreements**

Use the per-line CSV to find incorrect labels and default-speaker assignments. Verify the disputed speaker names by listening, and inspect whether a different assignment rule improves the same saved outputs.

2 **Broaden the test set**

Add customer-style interviews in our working languages. Precision-3 also leads line accuracy on the three recordings within NVIDIA's eight-speaker limit: 76.69% versus 68.53%.

3 **Measure correction work and cost**

Compare actual user corrections, latency and deployment cost. Independently reviewed speech boundaries can strengthen DER evaluation; no new inference is required to rescore cached results.

checksub.

Speaker detection evaluation · 2 November 2026

Five completed recordings · exploratory study